MedCLIP-SAM v1

Problem

-Negative-Positive-Coupling (NPC): when using small batch size (4k for 15M pairs)
-application of existed foundation models exist for data-efficient (zero-shot) medical image segmentation is limited and required more investigation

Evaluation & Results

-datasets and modalities: tested across three primary radiological modalities and tasks:
1-Breast Tumor Ultrasound
2-Brain Tumor MRI
3-Lung Chest X-ray
-Evaluation Metrics
1-Dice-Sorensen Coefficient (DSC): to evaluate the volumetric overlap between the predicted mask and the ground truth
2-Normalized Surface Distance (NSD): to assess the accuracy of the segmentation boundaries
-Results:
1-Zero-shot Setting: avg DSC of 64.54% and NSD of 66.10%
2-Weakly Supervised Setting: by refining zero-shot pseudo-labels through a secondary network the performance improved to avg DSC of 70.90% and NSD of 73.77%

Future work

-Extension to 3D/Volumetric Data
-Multi-class Segmentation
-Expanded Modalities: research will expand beyond radiology to include other biomedical imaging types such as histopathology and surgical videos
-reduce the reliance on manual inputs by allowing more intuitive natural language interactions (typed or spoken) and using LLMs like GPT-4 to automatically tailor prompts to specific clinical tasksf

Dataset

1-Breast Tumor Ultrasound: Breast Ultrasound Images (BUSI) for training and UDIAT for validation and testing, they contain ultrasound images of benign and malignant breast tumors (BUSI 600 images for training, UDIAT 50 images for validation and 113 images for testing)
2-Brain Tumor MRI: consists of T1-weighted MRI scans specifically for brain tumor segmentation, the MRI slices are provided at a resolution of 512*512 (training 1462, validation 1002, testing 600, total 3064)
3-Lung Chest X-ray: COVID-19 Radiography Database (COVID-QU-Ex) contains chest X-rays covering various cases including normal, lung opacity, viral pneumonia and COVID-19 with resolution of 299*299 (training 16280, validation 1372, testing 957, total 18609 images)

Training ptotocol

1-Foundation Model Initialization: instead of training a segmentation model from scratch, the framework utilizes pre-trained foundation models:
-BiomedCLIP: used as the backbone for multi-modal feature extraction, it consists of a vision encoder and a text encoder (PubMedBERT) that have been pre-trained on millions of biomedical image-text pairs to align medical concepts with visual features
-SAM (Segment-Anything-Model): used as the downstream segmentation engine
2-Zero-Shot Prompt Generation (the gScoreCAM stage): the training in the zero-shot stage does not involve updating weights on segmentation masks but rather generating guidance for SAM:
-Saliency Mapping: uses gScoreCAM to generate visual saliency maps, this technique decouples global textual information from the text encoder to identify which local visual regions in the image correspond to the text prompt
-Visual Prompts: these saliency maps are post-processed into bounding boxes, these boxes act as the prompts that tell SAM exactly where to perform segmentation
-Weakly Supervised Refinement: to improve accuracy beyond the zero-shot results, the protocol includes a weakly supervised learning phase:
-Pseudo-labeling: generated masks by the SAM zero-shot process are treated as pseudo-labels
-Secondary Training: supervised segmentation network (nnUNet) is trained from scratch using these pseudo-labels which allows the model to learn domain-specific features of the dataset (such as breast ultrasound or brain MRI) without requiring human-annotated ground truths

Proposed approach: MedCLIP-SAM

-gScoreCAM: works by leveraging the semantic alignment between text and vision in CLIP to create heatmaps that explain which parts of an image match a specific medical description
-MedCLIP-SAM v1 is a framework designed for universal, text-driven medical image segmentation that operates in both zero-shot and weakly supervised settings, bridges vision-language understanding with interactive segmentation by integrating three core components:
1-Multimodal Feature Extraction: utilizes BiomedCLIP as its foundation to extract and align features from medical images and text prompts with ViT and PubMedBERT, the model extracts global [EOS] and [CLS] tokens to represent text and images in a shared embedding space
2-Saliency-Guided Prompt Generation: employs a visual explanation technique to translate text prompts into a format that a segmentation model can understand (gScoreCAM and visual prompts where the saliency maps generated by gScoreCAM are post-processed to produce bounding boxes and these boxes serve as the visual prompts that indicate the target region of interest based on the text description
3-Interactive Segmentation with SAM: the bounding boxes generated in the previous step are fed into SAM which uses these prompts to perform the final dense pixel-level segmentation, producing a mask without requiring the user to manually draw on the image
4-Weakly Supervised Refinement: architecture includes a paradigm to further improve accuracy (Pseudo-labeling: masks produced by the zero-shot process are treated as pseudo-labels and Model Training: these pseudo-labels are used to train a secondary segmentation network to refine the results and enhance performance across specific datasets)

Limitation

-the v1 framework relied on gScoreCAM to generate visual prompts from text which was found to be less effective at capturing disease-relevant radiological representations compared to newer techniques leading to lower zero-shot accuracy
-v1 utilized the pre-trained BiomedCLIP model without the specialized DHN-NCE (Decoupled Hard Negative Noise Contrastive Estimation) loss, without this fine-tuning the model struggled to distinguish between subtle diagnostic imaging features within the same category
-the weakly supervised phase in v1 lacked interpretability, it didn't provide uncertainty maps which are crucial for clinical adoption to help users identify where a model's prediction might be unreliable
-v1 was validated on only three modalities (Breast Ultrasound, Brain MRI and Lung X-ray), it lacked the broader validation across more complex tasks like Lung CT segmentation
-v1 was limited to 2D inputs due to the underlying architecture of BiomedCLIP and focused strictly on binary segmentation (one class at a time) rather than multi-class clinical applications